Executive Summary

Predictive analytics and population health intelligence for HealthConnect Rwanda

Prepared by: Emmanuella Esi Arhin

Bottom line

The data cannot yet support clinical targeting, admission prediction or expansion decisions. It can, today, support three decisions: fix the data pipeline, recover unpaid revenue, and address low patient satisfaction as a network-wide problem.

What I analysed

27 datasets (273,602 rows) covering 20 facilities, 20,000 patients, 5,000 appointments, 60,000 bills and 21,064 feedback records. We cleaned and validated the data, ran descriptive, correlation and hypothesis analysis, built forecasts and predictive models, and segmented patients, facilities and departments.

Key findings

- **Data integrity is the main obstacle.** Only 5,000 of 60,000 bills, diagnoses and payments link to an appointment; the facility, department and doctor on an appointment disagree about 95% of the time; there are no admissions, lab, prescription or referral tables; every bed is recorded as available.
- **Money is being left uncollected.** 0.92bn RWF (9.9% of billings) is unpaid, and the Paid/Pending status on bills does not match what was actually paid. 21,028 insurance claims worth 2.44bn RWF are still pending.
- **Two in five patients are unhappy.** 40% of 21,064 ratings are 1-2. No facility or department is significantly better or worse, and no clinical, financial or operational factor predicts satisfaction.
- **Demand is flat and evenly spread.** No trend in appointments, diagnoses or revenue; no disease is increasing; no department or facility stands out significantly. Forecasts for 2027 are flat at the 2026 average (about 417 appointments a month in the extract).
- **Predictive models found no signal.** The admission-proxy model scores AUC 0.50 (0.50 is chance); satisfaction models do no better than guessing the average; age, sex and province do not identify who is at risk of chronic disease (75% prevalence everywhere).
- **Staffing is uniform while load is not.** Doctors see between 5 and 28 appointments (mean 14.3), and facility load per clinical staff member differs by about 32%, yet rosters are nearly identical (57-60 clinical staff per facility).

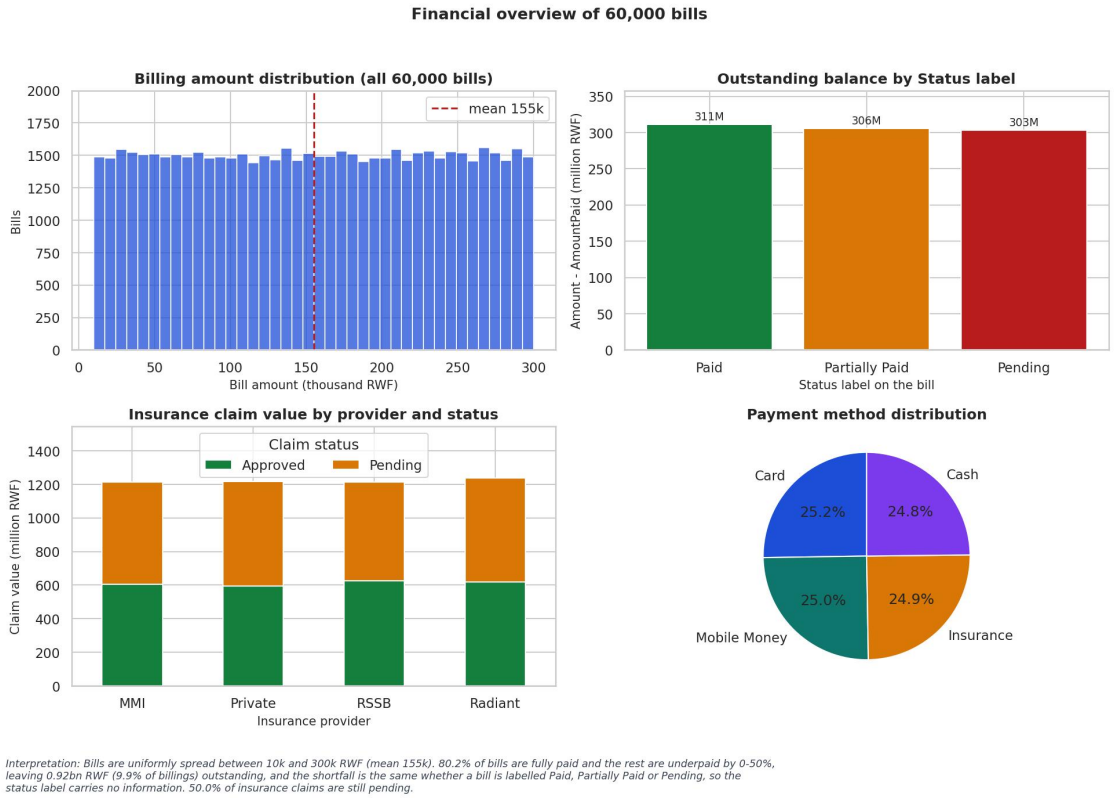


Figure: Financial overview of 60,000 bills

Bills are uniformly spread between 10k and 300k RWF (mean 155k). 80.2% of bills are fully paid and the rest are underpaid by 0-50%, leaving 0.92bn RWF (9.9% of billings) outstanding, and the shortfall is the same whether a bill is labelled Paid, Partially Paid or Pending, so the status label carries no information. 50.0% of insurance claims are still pending.

Recommendations

Area	Recommendation	Confidence
Operational challenges	Fix the data pipeline first: enforce appointment-ID foreign keys, derive facility from the department/doctor hierarchy, and add the missing admissions, lab, prescription, referral and facility-master tables before any further modelling.	High
Operational challenges	Record bed status in real time (occupied/cleaning/out of service) and capture admission and discharge timestamps.	High
High-risk patient groups	Audit chronic-condition coding (especially for children). Until validated, use a universal chronic-care outreach for flagged high-risk patients (age 65+ with a chronic condition, or any chronic condition with a severe diagnosis) rather than age/sex/region targeting.	Medium

Area	Recommendation	Confidence
Patient satisfaction	Treat poor satisfaction as a network-wide service problem, not a site problem: run a network-wide patient-journey review (waiting, communication, billing). Link every survey to its appointment, and replace free-text comments with structured reason codes.	Medium
Resource optimisation	Pilot a scheduling rule that balances doctor panels, and move 2-3 clinical staff from the lowest-load sites (HF004, HF005, HF007) to the highest-load sites (HF013, HF014, HF011); re-measure after one quarter before a permanent change.	Low-Medium
Revenue improvement	Rebuild payment status from payment records (Amount - AmountPaid), start a collections list for the largest underpaid bills, and set a claim turnaround target for all four insurers (e.g. clear pending claims within 30 days).	High
Revenue improvement	Reconcile finance reporting to the billing ledger monthly and explain the May, September and November spikes before using revenue forecasts for budgeting.	High
Expanding services	Do not commit capital to expansion on this evidence. Re-run the demand forecast once 24+ months of complete data exist, and use Cardiology and Radiology as the first capacity-review candidates.	Medium

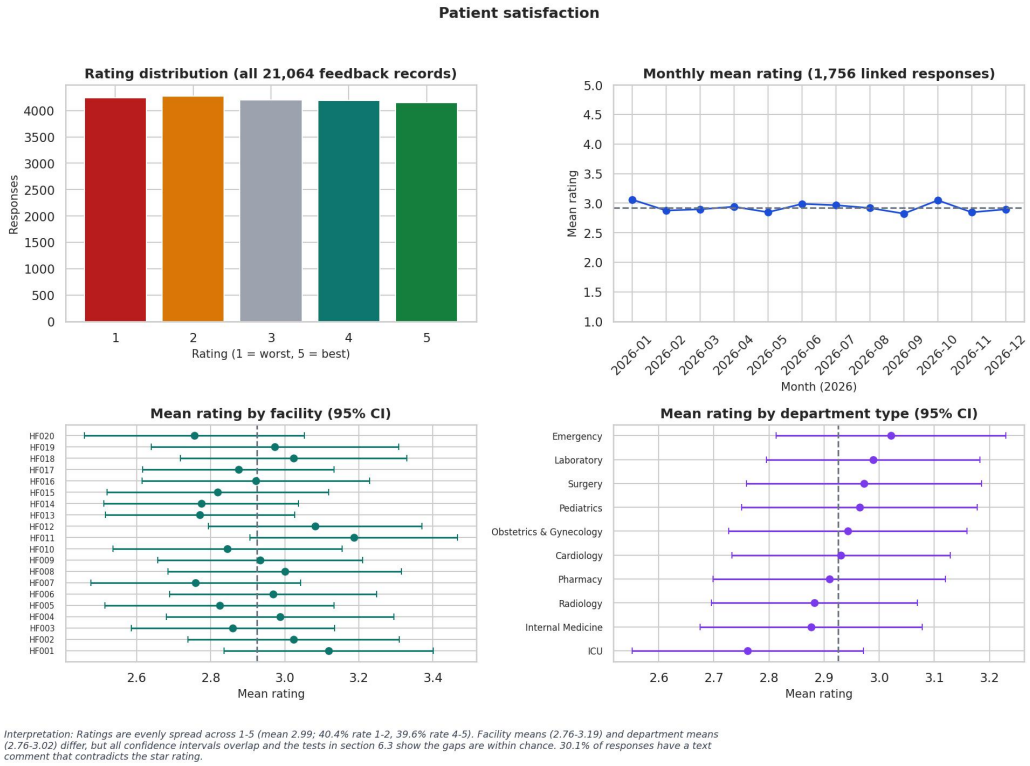


Figure: Patient satisfaction

Ratings are evenly spread across 1-5 (mean 2.99; 40.4% rate 1-2, 39.6% rate 4-5). Facility means (2.76-3.19) and department means (2.76-3.02) differ, but all confidence intervals overlap and the tests in section 6.3 show the gaps are within chance. 30.1% of responses have a text comment that contradicts the star rating.

Caveats and next steps

- Treat all results as provisional: the data show uniform distributions and no age, sex or severity gradients, which is unusual for real clinical records.
- Next 30 days: repair appointment keys, obtain the missing tables, rebuild payment status from payment records, and link every satisfaction survey to its appointment.
- Next quarter: re-run admission, length-of-stay and risk models on real outcomes, re-forecast with 24+ months of data, and add fairness checks by sex, age band and province before any deployment.